

Spotify Power BI Dashboard Using Advanced DAX Functions

Steps In Project

- ✓ Requirement Gathering / Business Requirements
- ✓ Data Walkthrough
- ✓ Data Connection
- ✓ Data Cleaning / Quality Check
- ✓ Data Modelling
- ✓ Data Processing
- ✓ DAX Calculations
- ✓ Dashboard Lay outing
- ✓ Charts Development and Formatting
- ✓ Dashboard / Report Development
- ✓ Insights Generation

Business Requirements

In today's digital music era, understanding listening patterns is crucial for both users and streaming platforms. This analysis focuses on Spotify Album Data, providing insights into user engagement with albums over time.

Albums:

1. Total Albumns Played over time: Track how album listening trends change over months and years.
2. Number of Album Listened by Year: Identify annual listening habits and volume.
3. Top 5 Albums: Identify the most played albums based on listening frequency.
4. Latest Year vs Previous Year Analysis: Compare album consumption between the latest and previous years including:
 - LY (Latest Year) vs PY (Previous Year) Trends
 - YoY (Year-over-Year) Growth Analysis

Listening Patterns

1. Listening Hours Analysis - Identify peak listening times using a Heat Map that visualizes patterns across hours and days with color intensity.
2. Average Listening Time (min) vs Track Frequency – Use a Scatter Plot with Quadrant Analysis to categorize tracks based on:
 - High Frequency & High Listening Time – Most engaging tracks
 - Low Frequency & High Listening Time – Niche but impactful tracks
 - High Frequency & Low Listening Time – Short & frequently played tracks
 - Low Frequency & Low Listening Time – Less popular tracks

DETAILS GRID

In this report, we aim to analyze Spotify data by creating an interactive and dynamic Grid View. The Grid will display key details such as Album Name, Artist Name, Track Name, and other relevant attributes.

Key Requirements:

1. Grid View with Essential Fields:

- The Grid should present critical data points for an intuitive and structured view.

2. Drill Through Functionality:

- Users should be able to drill through from the main reports to explore underlying data for detailed insights.
- The drilled-through data should be exportable to a CSV file based on user requirements.

3. Drill Down, Drill Up, and Hierarchy:

- Grid should support hierarchical navigation, allowing users to drill down and up for in-depth data exploration.

Domain Document

1. `spotify_track_uri`

- **Description:** A unique identifier assigned to each track in Spotify's database.
- **Format:** "spotify:track:<base-62 string>" (e.g., `spotify:track:3n3Ppam7vgaValiaRUc9Lp`).
- **Purpose:** Helps link tracks to their metadata and allows for cross-referencing with Spotify's catalog.

2. `ts` (Timestamp)

- **Description:** The exact time (in UTC) when the track stopped playing.
- **Format:** ISO 8601 format (e.g., `2024-02-07T14:30:45Z`).
- **Purpose:** Used for analyzing listening patterns, session durations, and track end times.

3. `platform`

- **Description:** The device or platform used to stream the track.
- **Possible Values:**
 - a) "desktop" (Windows/Mac app)
 - b) "mobile" (iOS/Android app)
 - c) "web" (Spotify Web Player)
 - d) "smart_speaker" (Amazon Echo, Google Home, etc.)
- **Purpose:** Helps understand where users are consuming music.

4. `ms_played`

- **Description:** The total number of milliseconds the track was played before stopping or skipping.
 - **Format:** Integer value (e.g., 215000 for 3 minutes 35 seconds).
 - **Purpose:** Useful for engagement analysis, identifying completed plays, and calculating revenue (as Spotify pays artists based on streaming duration).
5. track_name
- **Description:** The title of the song being played.
 - **Example:** "Shape of You"
 - **Purpose:** Helps in analyzing the most played tracks.
6. artist_name
- **Description:** The name of the artist performing the song.
 - **Example:** "Ed Sheeran"
 - **Purpose:** Useful for ranking popular artists and identifying user preferences.
7. album_name
- **Description:** The name of the album the track belongs to.
 - **Example:** "÷ (Divide)"
 - **Purpose:** Helps analyze album popularity and user listening trends.
8. reason_start
- **Description:** The reason why the track started playing.
 - **Possible Values:**
 - a) "trackdone" (Previous track ended)
 - b) "clickrow" (User manually selected the song)
 - c) "backbtn" (User pressed back)
 - d) "fwdbtn" (User pressed next)
 - e) "playbtn" (User pressed play)
 - f) "autoplay" (Spotify automatically selected the next track)
 - **Purpose:** Helps understand user behavior and track engagement patterns.
9. reason_end
- **Description:** The reason why the track stopped playing.
 - **Possible Values:**
 - a) "trackdone" (Track finished playing)
 - b) "endplay" (User paused or stopped playback)
 - c) "fwdbtn" (User skipped to the next track)
 - d) "backbtn" (User went back to the previous track)
 - e) "logout" (User logged out or session ended)
 - **Purpose:** Helps identify why users stop listening to tracks, which is crucial for retention analysis.
10. shuffle
- **Description:** Indicates whether shuffle mode was enabled during playback.
 - **Possible Values:**
 - a) TRUE (Shuffle mode was ON)
 - b) FALSE (Shuffle mode was OFF)
 - **Purpose:** Helps analyze how often users use shuffle mode in their listening sessions.

11. skipped

- **Description:** Indicates whether the user skipped the song before it finished.
- **Possible Values:**
 - a) `TRUE` (User skipped to the next track)
 - b) `FALSE` (User did not skip)
- **Purpose:** Important for understanding user engagement, drop-off rates, and song popularity.